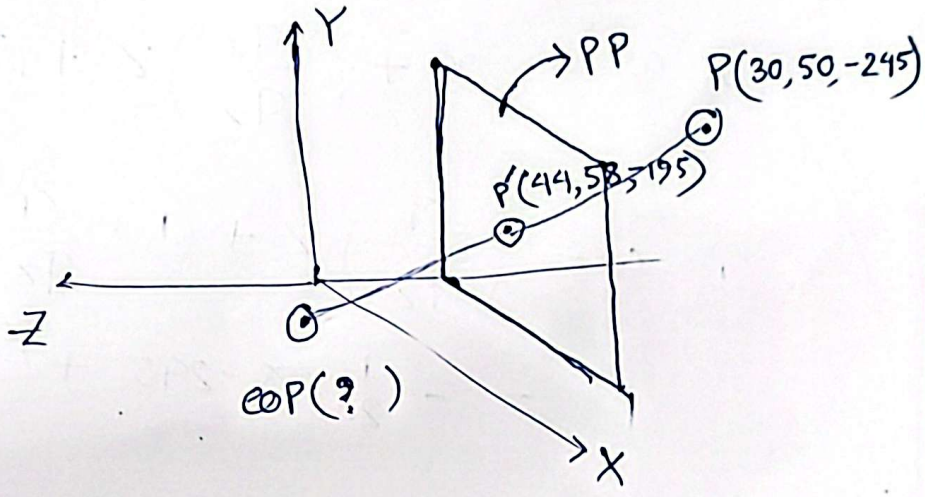
CSE 423 Quiz - 5

Set: A

Mark: 20, Time: 20 minutes

Name: ID: Section:

1. The projected point P' (44, 58, -195) for a point P (30, 50, -245) is given. The Projection Plane (PP) (XY plane parallel) is 200 units away from COP along Z axis. Find the COP.



Projected point $P'(44, 58, -195)$ lies on the Projection plane.

As the projection plane is XY Parallel ($Z = d$ type)

$\therefore$ The projection plane is $Z_p = -195$ (Z value of P')

Let, $COP = (COP_x, COP_y, COP_z)$

$- \quad (0, 0, Z_p) = (0, 0, -195)$

$(q_x, q_y, q_z) = (COP_x, COP_y, COP_z + 195)$

COP is 200 unit from PP. (in the positive direction. Why? PP lies between COP and P)

$\therefore COP_z = -195 + 200 = 5$

$$\therefore (q_x, q_y, q_z) = (cop_x, cop_y, 200)$$

$P'(44, 58, -195)$ $\quad$ $P(30, 50, -245)$

x', y', z' $\quad$ x, y, z

$$x'_\omega = x + \left(\frac{-q_x}{q_z}\right) z + \left(z_p \cdot \frac{q_x}{q_z}\right)$$

$$\omega = \left(\frac{-1}{q_z}\right) z + \left(1 + \frac{z_p}{q_z}\right)$$

$$= -\frac{1}{200} \times -245 + \left(1 + \frac{-195}{200}\right)$$

$$= 1.25$$

$$\rightarrow 44 * 1.25 = 30 + \frac{-q_x}{200} \times -245 + \left(-195 \times \frac{q_x}{200}\right)$$

$$\Rightarrow q_x = 100 = cop_x$$

$$y'_\omega = y + \left(\frac{-q_y}{q_z}\right) z + \left(z_p \frac{q_y}{q_z}\right)$$

$$\Rightarrow 58 \times 1.25 = 50 + \frac{-q_y}{200} \times -245 + \left(-195 \times \frac{q_y}{200}\right)$$

$$q_y = 90 = cop_y$$

$$(cop_x, cop_y, cop_z) = (100, 90, 5)$$

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